

Measuring a Planet's Radius from Horizon Curvature

An exact two-ruler method

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Abstract

The visible horizon of a spherical planet is depressed below the observer's local horizontal plane and, projected onto a plane held in front of the eye, is curved. Both effects depend on the planet's radius. This note gives an exact geometric inversion of that dependence using two ordinary rulers fixed in a single vertical plane. The measured quantities are the eye altitude h above the spherical surface, the perpendicular distance D from the eye to the ruler plane, the separation L of the two points at which the horizon crosses a horizontal ruler, and the rise s of the horizon above that ruler at the midpoint of the crossings, read from a second, vertical ruler. The projected horizon is shown to be one branch of a hyperbola, from which the exact horizon depression angle and then the planet radius $R = h(\sqrt{1 + T^2} + 1)/T^2$ follow in closed form. No small-angle expansion, circular-arc substitution, or other approximation is used at any step; where a numerically stable rearrangement is given, it is algebraically identical to the form it replaces. The sensitivity of the method and its systematic limitations are discussed, and the result is compared with earlier horizon-curvature measurements.

1 Introduction

The horizon seen from a spherical planet lies below the observer's local horizontal plane, and its image on a plane held in front of the eye is curved rather than straight. Both the depression and the curvature depend on the planet's radius and on the observer's altitude, so a sufficiently careful observation of the horizon determines the radius. The depression, or dip, is classical and is used in celestial navigation; its unrefracted and refracted forms are reviewed by [Young \(n.d.\)](#). The curvature is the harder of the two effects to see, because from ordinary altitudes the horizon departs from a straight line by far less than the width of a normal field of view, which is why photographic tests of it demand care, as [Lynch \(2008\)](#) discusses.

This note treats the inverse problem in its simplest instrumented form. Two ordinary rulers, held in one fixed vertical plane in front of a single eye, are enough to determine the planet radius R , provided the observer also knows the eye altitude above the spherical surface and the distance from the eye to the ruler plane. A horizontal ruler is placed so that the visible horizon crosses it near its two ends; a vertical ruler at the midpoint of those crossings reads the amount by which the horizon bows above the horizontal ruler. Four lengths are then in hand, and the radius follows in closed form.

Inferring a radius from the horizon is an old idea. In the eleventh century al-Bīrūnī measured the dip of the sea horizon from a hill of known height at Nandana, in present-day Pakistan, and inverted it for the Earth's radius, arriving at a figure within about two percent of the modern one ([Sparavigna 2013](#)); the inversion he used is equation (8) below. Measuring the dip directly then became an instrument problem for navigators, who built dedicated dip meters and horizon meters for it, and who found the observed dip

departing from tabulated values by ten minutes of arc and more under anomalous refraction (Moll 1931). The present method reaches the same dip angle through two ruler readings instead of an angle-measuring instrument.

Measurements of the horizon's curvature have been reported more recently. Lynch (2008) extracted the analogous displacement from three horizon points in a photograph; Kinlen (2017) derived the projection of the horizon onto a vertical image plane; Howe (2019) described a closely related comparison of the horizon against a straight edge in a photograph; and Cyrulies (2025) built a dedicated instrument with a rotating sight, reticle, and camera; and Bislin (n.d.) models the same midpoint displacement numerically, with refraction, in a widely used curvature simulator. Earlier still, Lichtenstein and Suit (1967) tested calibrated horizon templates for the inverse problem of estimating altitude above a body of known radius. Section 8 compares these with the arrangement used here.

The emphasis throughout is on exactness. No step uses $\sin x \approx x$, $\tan x \approx x$, $\cos x \approx 1 - x^2/2$, a circular arc in place of the true projected curve, or any approximation justified by the size of the planet or the smallness of the measured rise. Where an algebraically equivalent rearrangement is introduced for numerical stability, it is identified as such. Section 2 defines the four measured quantities and the observing arrangement, Section 3 derives the exact geometry, Section 4 inverts it for the radius, Section 5 states the procedure as a checklist, Section 6 works a numerical example, and Sections 7 and 8 discuss accuracy and prior work.

2 The Measurement

2.1 Measured quantities

The method needs four measured lengths:

- h altitude of the observer's eye above the local spherical surface;
- D perpendicular distance from the eye to the ruler plane;
- L horizontal distance between the two points where the horizon crosses the horizontal ruler;
- s vertical distance from the horizontal ruler to the horizon at the midpoint of the two crossings.

The altitude h must be measured above the same spherical surface that forms the horizon. For a sea horizon viewed from a hill, it therefore includes the hill's elevation above sea level as well as the eye's height above the ground. All four lengths must be expressed in the same unit.

Throughout, assume $R > 0$, $h > 0$, $D > 0$, and $L > 0$. The arrangement described below then gives $s > 0$, as confirmed in Section 3.3.

The distance D is essential and cannot be dispensed with. Without it, L and s describe only a shape drawn on the ruler, not the angular size of that shape as seen from the eye. Different eye-to-ruler distances can produce the same L and s while corresponding to different planet radii.

2.2 Observation setup

Hold the horizontal ruler in a vertical plane in front of the eye, with the plane perpendicular to the central horizontal line of sight, and position the ruler so that the visible horizon crosses it near its left and right edges. Use one fixed eye, and place both rulers in the same vertical plane. The vertical ruler must lie at the midpoint of the two horizon crossings, directly in front of the eye in the left-right direction. Measure D horizontally and perpendicularly to that plane, not along the downward sightline to the horizon.

At the centre of the horizontal ruler, use the vertical ruler to measure the distance from the horizontal ruler to the visible horizon. In the coordinate convention of Section 3.1, s is positive when the horizon at the midpoint stands above the horizontal ruler, which is the expected arrangement whenever the horizontal ruler passes through two symmetric horizon points. The arrangement is shown in Figure 1.

Howe (2019) describes a closely related ruler comparison carried out on a photograph. Here the second ruler reads the midpoint rise directly, in the same plane and at the same moment as the crossings.

The measurement is best made with the head and rulers fixed, and with the horizontal ruler levelled relative to the local horizontal direction. The derivation assumes that the visible horizon is the true geometric horizon of a spherical planet, and not a horizon displaced by hills, waves, buildings, haze, or atmospheric refraction.

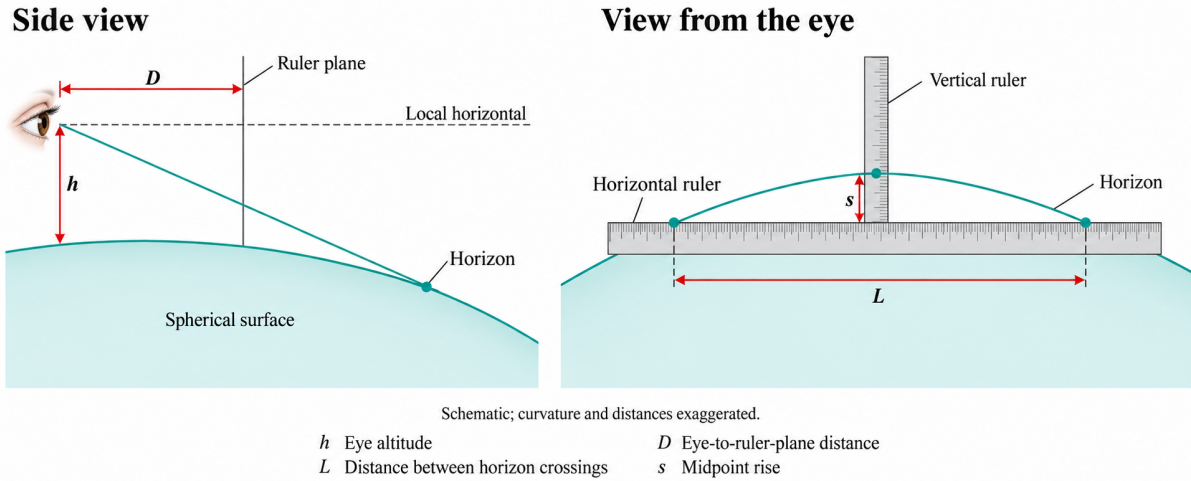


Figure 1. Observation setup. Left: eye altitude h and perpendicular eye-to-ruler-plane distance D . Right: the observer's view, with crossing separation L and midpoint rise s measured from the horizontal ruler's top edge. The drawing is schematic; curvature and distances are exaggerated.

3 Exact Horizon Geometry

3.1 Coordinates and conventions

Place the observer's eye at the origin $O = (0, 0, 0)$ of a Cartesian frame with x the horizontal left-right direction along the ruler, y the horizontal forward direction from the eye toward the ruler plane, and z the vertical upward direction. The planet centre then lies directly below the eye at

$$C = (0, 0, -(R + h)),$$

where R is the unknown planet radius, and the ruler plane is the plane $y = D$. Write $a = L/2$ for the half-width of the crossings.

In this frame the two measured horizon crossings lie at $(\pm a, D, z_e)$ and the horizon point directly in front of the eye lies at $(0, D, z_0)$. Both heights are negative, because the visible horizon is below eye level, while the measured midpoint rise

$$s = z_0 - z_e \tag{1}$$

is positive. Signs should be checked against (1) whenever the prose or the equations are changed.

3.2 Depression angle of the geometric horizon

A ray leaving the eye in the unit direction $\mathbf{u}$, with $|\mathbf{u}| = 1$, reaches the geometric horizon when the line it defines is tangent to the planet, that is, when the perpendicular distance from the planet centre C to that line equals R . That distance is $\sqrt{|C|^2 - (C \cdot \mathbf{u})^2}$, so the tangency condition is

$$|C|^2 - (C \cdot \mathbf{u})^2 = R^2.$$

Since $|C| = R + h$ and $C \cdot \mathbf{u} = -(R + h)u_z$, this becomes $(R + h)^2(1 - u_z^2) = R^2$, and dividing by $(R + h)^2$,

$$1 - u_z^2 = \left(\frac{R}{R + h} \right)^2.$$

For the visible horizon, which lies below eye level, u_z is negative, so

$$u_z = -\sqrt{1 - \left(\frac{R}{R + h} \right)^2}.$$

For this root $C \cdot \mathbf{u} = -(R + h)u_z > 0$, so the tangent point lies on the forward ray from the eye and not on its backward extension.

Define the exact depression angle δ of the horizon below the local horizontal plane, with $0 < \delta < \pi/2$, by $u_z = -\sin \delta$. Then $\sin \delta = \sqrt{1 - (R/(R + h))^2}$, and therefore

$$\cos \delta = \frac{R}{R + h}. \quad (2)$$

Equation (2) is exact for a spherical planet with straight light rays. It is the standard unrefracted dip relation, given in the same form by [Young \(n.d.\)](#).

3.3 Projection onto the ruler plane

A point (x, D, z) of the ruler plane is seen from the eye along the vector $\mathbf{v} = (x, D, z)$, whose unit direction $\mathbf{u} = (x, D, z)/\sqrt{x^2 + D^2 + z^2}$ has vertical component $u_z = z/\sqrt{x^2 + D^2 + z^2}$. The point lies on the projected horizon exactly when this component equals $-\sin \delta$:

$$\frac{z}{\sqrt{x^2 + D^2 + z^2}} = -\sin \delta. \quad (3)$$

Squaring (3) gives $z^2 = \sin^2 \delta (x^2 + D^2 + z^2)$, that is, $z^2(1 - \sin^2 \delta) = \sin^2 \delta (x^2 + D^2)$. The exact identity $1 - \sin^2 \delta = \cos^2 \delta$ and division by $\cos^2 \delta > 0$ turn this into $z^2 = \tan^2 \delta (x^2 + D^2)$. The visible horizon is below the local horizontal direction, so $z < 0$ and

$$z(x) = -\tan \delta \sqrt{x^2 + D^2}. \quad (4)$$

The positive root discarded here is the mirror image of the horizon above eye level, which does not satisfy (3) and is excluded by the sign of u_z . Equation (4) is the lower branch of the hyperbola $z^2 - x^2 \tan^2 \delta = D^2 \tan^2 \delta$, and it is the exact projected horizon curve on the vertical ruler plane. It is not replaced here by a circular arc. [Kinlen \(2017\)](#) derives the equivalent projection for a vertical image plane, his viewing

distance v corresponding to D and his midpoint rise ϵ to s ; the two rulers supply a physical measurement in that plane.

Evaluating (4) at the midpoint $x = 0$ gives $z_0 = -D \tan \delta$, and at either crossing $x = \pm a$ it gives $z_e = -\tan \delta \sqrt{a^2 + D^2}$. Substituting both into (1) and factoring out $\tan \delta$,

$$s = \tan \delta (\sqrt{a^2 + D^2} - D). \quad (5)$$

Because $\sqrt{a^2 + D^2} > D$ for $a > 0$, equation (5) confirms that $s > 0$ under the stated assumptions. Lynch (2008, Section 3 and Figure 5) measures the analogous displacement from three horizon points in a photograph; the circular-sagitta model he introduces later is not the exact hyperbolic projection derived here.

4 Inversion for the Planet Radius

4.1 The depression angle from the ruler readings

Solving (5) for the tangent of the depression angle and using $a = L/2$,

$$\tan \delta = \frac{s}{\sqrt{a^2 + D^2} - D} = \frac{s}{\sqrt{D^2 + (L/2)^2} - D}. \quad (6)$$

For numerical evaluation the denominator of (6) can be rationalized exactly. Since

$$(\sqrt{D^2 + a^2} - D)(\sqrt{D^2 + a^2} + D) = (D^2 + a^2) - D^2 = a^2,$$

multiplying numerator and denominator by $\sqrt{D^2 + a^2} + D$ gives the quantity

$$T \equiv \tan \delta = \frac{s(\sqrt{D^2 + a^2} + D)}{a^2} = \frac{4s(\sqrt{D^2 + (L/2)^2} + D)}{L^2}. \quad (7)$$

Equation (7) avoids subtracting two nearly equal numbers when L is small relative to D , which is the usual case. It is algebraically identical to (6) and is not an approximation.

4.2 The radius

The dip relation (2) inverts directly. Multiplying out $\cos \delta (R + h) = R$ gives $R \cos \delta + h \cos \delta = R$, hence $h \cos \delta = R(1 - \cos \delta)$, and dividing by $1 - \cos \delta$,

$$R = \frac{h \cos \delta}{1 - \cos \delta}. \quad (8)$$

Equation (8) is al-Bīrūnī's result: it turns a dip angle observed from a known height into the planet radius, and everything before it in this article exists only to supply δ from two ruler readings.

Since $0 < \delta < \pi/2$, the cosine is positive and the exact identity $\cos \delta = 1/\sqrt{1 + \tan^2 \delta}$ applies, so $\cos \delta = 1/\sqrt{1 + T^2}$ with T from (7). Substituting into (8) and multiplying numerator and denominator by $\sqrt{1 + T^2}$,

$$R = \frac{h/\sqrt{1 + T^2}}{1 - 1/\sqrt{1 + T^2}} = \frac{h}{\sqrt{1 + T^2} - 1}. \quad (9)$$

Rationalizing (9) in the same way as before gives the form that should be used in practice,

$$R = \frac{h(\sqrt{1 + T^2} + 1)}{T^2}, \quad (10)$$

which avoids subtracting two nearly equal numbers when T is small. Like (7), it is an exact rearrangement and not an approximation.

4.3 Result

Measuring h , D , L , and s in a common unit and combining (7) with (10) gives the planet radius as

$$T = \frac{4s(\sqrt{D^2 + (L/2)^2} + D)}{L^2}, \quad R = \frac{h(\sqrt{1 + T^2} + 1)}{T^2}.$$

Two degenerate cases bound the method's applicability. At $h = 0$ the geometric horizon gives $s = 0$ for every radius, so no observation at eye level on the surface can determine R . For $h > 0$, a measured $s \leq 0$ is incompatible with a finite positive radius in the stated arrangement and must not be inserted into (10), where squaring T would conceal the wrong sign.

5 Procedure

1. Measure the eye altitude h above the spherical surface that forms the horizon.
2. Fix both rulers in one vertical plane at the measured horizontal, perpendicular distance D from the observing eye.
3. Level the horizontal ruler against the local horizontal direction.
4. Align the horizontal ruler so that the visible horizon crosses it at two symmetric points near its edges, with their midpoint directly in front of the eye in the left-right direction.
5. Measure the distance L between those two crossings.
6. At the midpoint of the crossings, measure the vertical rise s from the horizontal ruler to the horizon.
7. Compute $T = 4s(\sqrt{D^2 + (L/2)^2} + D)/L^2$.
8. Compute $R = h(\sqrt{1 + T^2} + 1)/T^2$.

6 Worked Example

The numbers in this section are synthetic. The rise s was computed from the forward relation (5) for an assumed radius $R = 6371.0$ km, the mean radius of the Earth, and then rounded to the nearest 0.001 mm to imitate a ruler reading. The example therefore checks the inversion and exhibits its sensitivity; it is not the report of an observation, and by construction it contains no refraction, no wave or terrain displacement of the horizon, and no alignment error.

Take an observer on a coastal summit looking out at the sea horizon, with a metre rule held at arm's length and a second rule standing at its midpoint:

Quantity	Symbol	Value
Eye altitude above the sea surface	h	2000.0 m
Eye to ruler plane	D	0.6000 m
Separation of the horizon crossings	L	0.6000 m
Midpoint rise	s	1.775 mm = 1.775×10^{-3} m

Quantity	Symbol	Value
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All four are already in metres, as (7) and (10) require. Evaluating the bracket in (7) first,

$$\sqrt{D^2 + (L/2)^2} = 0.6708204 \text{ m}, \quad \sqrt{D^2 + (L/2)^2} + D = 1.2708204 \text{ m}, \quad L^2 = 0.360000 \text{ m}^2,$$

so that

$$T = \frac{4 (1.775 \times 10^{-3}) (1.2708204)}{0.360000} = 2.506340 \times 10^{-2},$$

which corresponds to a horizon depression of $\delta = \arctan T = 1.4357^\circ$. Then $T^2 = 6.281741 \times 10^{-4}$ and $\sqrt{1 + T^2} = 1.0003140$, so (10) gives

$$R = \frac{2000.0 (2.0003140)}{6.281741 \times 10^{-4}} = 6.368661 \times 10^6 \text{ m} = 6368.7 \text{ km}.$$

This is 0.037% below the assumed 6371.0 km, and the entire discrepancy comes from rounding s to the nearest 0.001 mm. The example also shows why (7) and (10) are preferred over their unrationalized equivalents: here $\sqrt{1 + T^2} - 1 = 3.1404 \times 10^{-4}$, so the denominator of (9) is the difference of two numbers that agree to three decimal places, and the agreement tightens as T falls.

The sensitivity of the result to the ruler reading can be stated exactly. Differentiating (9) and using $R = h/(\sqrt{1 + T^2} - 1)$ to eliminate h ,

$$\frac{\partial \ln R}{\partial \ln T} = - \frac{T^2}{\sqrt{1 + T^2} (\sqrt{1 + T^2} - 1)}, \quad (11)$$

and T is exactly proportional to s at fixed D and L , so $\partial \ln R / \partial \ln s$ takes the same value. At the T of this example (11) equals -1.99969 : the inferred radius responds twice as strongly as the reading, and in the opposite direction. The tendency of (11) to -2 as $T \rightarrow 0$ is a limit, not an approximation. Its practical weight is easiest to see by recomputing the radius exactly from neighbouring readings. A rule read to ± 0.05 mm, which is 2.8% of s here, admits

$$s = 1.725 \text{ mm} \Rightarrow R = 6743 \text{ km}, \quad s = 1.825 \text{ mm} \Rightarrow R = 6025 \text{ km},$$

a spread of about $\pm 6\%$. By contrast $\partial \ln R / \partial \ln h = 1$ exactly, so the altitude enters only in proportion. Since s grows with L at fixed D and δ , widening the horizontal ruler relative to the eye-to-ruler distance is the most direct way to reduce the relative error, and it is why this example uses $L = D$.

7 Accuracy and Systematic Effects

The measured value of s is usually small. For an Earth-sized planet and a ruler held at ordinary arm's length, the midpoint rise can be a fraction of a millimetre unless the observer is at a large altitude. The method is therefore sensitive to ruler alignment, eye position, terrain, sea state, and atmospheric refraction.

Equations (7) and (10) describe an ideal spherical planet with a sharp geometric horizon. Real measurements should record the uncertainty in h , D , L , and s , and should repeat the observation with different ruler widths and eye-to-ruler distances. A valid result gives the same R within the combined uncertainty.

Equation (11) fixes how an error in the ruler reading propagates, and Section 6 shows what that amounts to for a plausible arrangement.

Refraction is the dominant systematic effect, and it is not small. With the standard refraction coefficient $k \approx 1/7$ near sea level (Young, n.d.), a horizon seen through the real atmosphere behaves like the horizon of a larger sphere, and a radius inferred from the geometric relations above is too large by roughly a factor $1/(1 - k)$, which is some seventeen per cent for $k = 1/7$. The bias is not stable either: Moll (1931) reports observed dips departing from tabulated values by ten minutes of arc and more, which at the altitudes available to this method is a large fraction of δ itself. That al-Bīrūnī's own result nevertheless came out within a few percent of the modern radius should not be read as evidence that the bias is small; agreement with a known answer is not a check on refraction.

Agreement between repeated measurements is a consistency check, not a proof that refraction is absent. A refracted horizon can shift the inferred radius systematically even when different ruler settings agree with one another. For the physical distinction between the geometric and the refracted horizon dip, see Young (n.d.).

8 Relation to Earlier Work

Cyruities (2025) reports a horizon-curvature experiment from a known altitude using a rotating sight, a reticle, and a camera; its instrument and its angular measurements differ from the fixed rulers used here. Earlier, Lichtenstein and Suit (1967) tested calibrated horizon templates to estimate altitude above a body of known radius, which is the related inverse problem. An informal proposal by skrutnizer (2022) also suggests estimating the radius from altitude and the angular bow of the horizon over a ruler; it is evidence of an earlier discussion of the idea, not a source for the equations above.

The method also sits in the older line of dip measurements. Al-Bīrūnī's inversion (8) is unchanged here; what differs is how δ is obtained. Navigational dip meters read it directly with a prism and a sextant scale (Moll 1931), whereas the two rulers infer it from lengths, which removes the need for a calibrated angular scale at the cost of requiring D . Closest in observable is Bislin (n.d.), whose curvature simulator computes the drop of the horizon relative to the chord joining two horizon points near the edges of a frame – the quantity called s here – including refraction. That model runs forward from an assumed radius and treats the horizon arc as circular, rather than inverting the exact projection (4) for R .

These precedents establish both the measurement principle and the projected geometry. What is presented here is an explicit two-ruler implementation together with an exact inversion in terms of h , D , L , and s . A search of the literature did not locate that complete combination, but a search that fails to find an arrangement does not establish its historical originality. Detailed comparisons are recorded in `notes/prior-work.md`.

9 Conclusion

Two rulers in a fixed vertical plane, a known eye altitude, and a known eye-to-ruler distance determine the radius of a spherical planet exactly, through equations (7) and (10). The projected horizon is a hyperbola rather than a circular arc, and the inversion requires no small-angle expansion at any stage. The practical difficulty lies entirely in the measurement: the midpoint rise s is small at accessible altitudes, and atmospheric refraction, which is outside the ideal model, biases the result in a way that repeated measurements alone cannot reveal.

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